Diag Solutions

---

1. **1.** The total surface area of a cube whose edge is 5 cm is:

   Answer: (A)

   Solution:

   1. State the formula for the total surface area of a cube: $TSA = 6a^2$.
   2. Substitute $a = 5$ cm.
   3. Calculate the final value.

   Derivation:

   $$TSA = 6 \times (5)^2 = 6 \times 25 = 150 \text{ cm}^2$$

   Surface Areas and Volumes — Surface area of cube

2. **2.** If the slant height of a cone is 10 cm and the radius is 6 cm, its curved surface area is:

   Answer: (D)

   Solution:

   1. State the formula for the curved surface area of a cone: $CSA = \pi rl$.
   2. Substitute $r = 6$ cm, $l = 10$ cm.
   3. Calculate the final value in terms of .

   Derivation:

   $$CSA = \pi \times 6 \times 10 = 60\pi \text{ cm}^2$$

   Surface Areas and Volumes — Curved surface area of cone

3. **3.** The lateral surface area of a cuboid of dimensions 8 cm $\times$ 6 cm $\times$ 5 cm is:

   Answer: (C)

   Solution:

   1. State the formula for the lateral surface area of a cuboid: $LSA = 2h(l + b)$.
   2. Identify $l = 8$ cm, $b = 6$ cm, $h = 5$ cm.
   3. Calculate the final value.

   Derivation:

   $$LSA = 2 \times 5 \times (8 + 6) = 10 \times 14 = 140 \text{ cm}^2$$

   Surface Areas and Volumes — Surface area of cuboid

4. **4.** Assertion (A): The total surface area of a solid hemisphere of radius r is $3\pi r^2$. Reason (R): The total surface area of a sphere is $4\pi r^2$.

Answer: (B)

Solution:

1. The total surface area of a hemisphere includes the curved surface area ($2\pi r^2$) and the area of the circular base (πr^2). Thus, TSA = $3\pi r^2$. Assertion is true.
2. The formula for the total surface area of a sphere is indeed $4\pi r^2$. Reason is true.
3. The reason explains the surface area of a sphere, not the derivation of the hemisphere's area. Thus, R is not the correct explanation for A.

Derivation:

$$TSA_{\text{hemisphere}} = 2\pi r^2 + \pi r^2 = 3\pi r^2$$

Surface Areas and Volumes — Surface Area of Solids

5. **5.** Assertion (A): If the radius of a cylinder is doubled and its height is halved, the volume remains the same. Reason (R): The volume of a cylinder is given by $V = \pi r^2 h$.

Answer: (D)

Solution:

1. Let initial volume be $V = \pi r^2 h$.
2. New volume $V' = \pi(2r)^2(h/2) = \pi(4r^2)(h/2) = 2\pi r^2 h = 2V$. Assertion is false.
3. The volume formula for a cylinder is correctly identified as $\pi r^2 h$. Reason is true.

Derivation:

$$V_{new} = \pi(2r)^2 \times \frac{h}{2} = 2\pi r^2 h$$

Surface Areas and Volumes — Volume of Cylinder

6. **6.** Assertion (A): The volume of a cone with radius r and height h is $\frac{1}{3}\pi r^2 h$. Reason (R): A cone is one-third of a cylinder with the same base and height.

Answer: (A)

Solution:

1. The volume of a cone is mathematically defined as $\frac{1}{3}\pi r^2 h$. Assertion is true.
2. Geometrically, a cone occupies one-third the volume of a cylinder having the same radius and height. Reason is true and explains the assertion.

Derivation:

$$V_{\text{cone}} = \frac{1}{3}V_{\text{cylinder}} = \frac{1}{3}\pi r^2 h$$

Surface Areas and Volumes — Volume of Cone

7. **7.** Assertion (A): The slant height l of a cone is given by $l = \sqrt{h^2 + r^2}$. Reason (R): This is derived from Pythagoras theorem applied to the triangle formed by height, radius, and slant height.

   Answer: (A)

   Solution:

   1. The slant height l, radius r, and vertical height h form a right-angled triangle in a cone.
   2. By Pythagoras theorem, the hypotenuse $l^2 = r^2 + h^2$, so $l = \sqrt{r^2 + h^2}$. Assertion is true.
   3. The Reason correctly identifies the geometric origin of the formula. Thus, R is the correct explanation.

   Derivation:

$$\begin{aligned} l^2 &= r^2 + h^2 \\ \implies l &= \sqrt{r^2 + h^2} \end{aligned}$$

   Surface Areas and Volumes — Cone Properties

8. **8.** Assertion (A): If the surface area of a sphere is 616 cm^2, its radius is 7 cm. Reason (R): The surface area of a sphere is $4\pi r^2$.

   Answer: (A)

   Solution:

   1. Using $4\pi r^2 = 616$, we have $4 \times (22/7) \times r^2 = 616$.
   2. $r^2 = 616 \times 7/(4 \times 22) = 616 \times 7/88 = 7 \times 7 = 49$.
   3. $r = 7$ cm. Assertion is true.
   4. Reason is the correct formula used for the calculation. Thus, R is the correct explanation.

   Derivation:

$$\begin{aligned} 4 \times \frac{22}{7} \times r^2 &= 616 \\ \implies r^2 &= 49 \\ \implies r &= 7 \end{aligned}$$

   Surface Areas and Volumes — Surface Area of Sphere

9. **9.** Two cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.

   Answer:

$$160 \text{ cm}^2$$

   Solution:

1. Determine the side of the cube using the volume formula $a^3 = 64$.
2. Identify the dimensions of the new cuboid ($l = 8, b = 4, h = 4$) and calculate the surface area using $2(lb + bh + lh)$.

Derivation:

$a = \sqrt[3]{64} = 4$ cm. New dimensions: $l = 8, b = 4, h = 4$. SA $= 2(8 \times 4 + 4 \times 4 + 8 \times 4) = 2(32 + 16 + 32) = 2(80) = 160$ cm^2.

Surface Areas and Volumes — Combination of solids

10. **10.** A metallic sphere of radius 4.2 cm is melted and recast into the shape of a cylinder of radius 6 cm. Find the height of the cylinder.

Answer:

$$2.744 \text{ cm}$$

Solution:

1. Equate the volume of the sphere to the volume of the cylinder: $\frac{4}{3}\pi r_1^3 = \pi r_2^2 h$.
2. Solve for h by substituting the given values.

Derivation:

$$\frac{4}{3}\pi(4.2)^3 = \pi(6)^2 h$$
$$\Rightarrow h = \frac{4 \times 74.088}{3 \times 36} = 2.744 \text{ cm.}$$

Surface Areas and Volumes — Conversion of solids

11. **11.** The slant height of a frustum of a cone is 4 cm and the perimeters of its circular ends are 18 cm and 6 cm. Find the curved surface area of the frustum.

Answer:

$$48 \text{ cm}^2$$

Solution:

1. Use the formula for curved surface area of a frustum: $CSA = \pi(r_1 + r_2)l$.
2. Since $2\pi r_1 = 18$ and $2\pi r_2 = 6$, note that $\pi(r_1 + r_2) = \frac{18+6}{2} = 12$.

Derivation:

$$\pi(r_1 + r_2) = \frac{18 + 6}{2} = 12. \text{ CSA} = 12 \times 4 = 48 \text{ cm}^2.$$

Surface Areas and Volumes — Frustum of a cone

12. **12.** A solid is in the form of a right circular cone mounted on a hemisphere. If the radius of the hemisphere and the cone is 3.5 cm and the height of the cone is 4 cm, find the volume of the solid. (Use $\pi = 22/7$)

Answer:

$$154 \text{ cm}^3$$

Solution:

1. Volume of solid = Volume of cone + Volume of hemisphere.
2. Apply $V = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$.

Derivation:

$$V = \frac{1}{3} \times \frac{22}{7} \times (3.5)^2 \times 4 + \frac{2}{3} \times \frac{22}{7} \times (3.5)^3 = 51.33 + 89.83 \approx 154 \text{ cm}^3.$$

Surface Areas and Volumes — Combination of solids

13. **13.** The diameter of a metallic ball is 4.2 cm. What is the mass of the ball, if the density of the metal is 8.9 g/cm^3?

Answer:

$$345.39 \text{ g}$$

Solution:

1. Calculate the volume of the sphere using $V = \frac{4}{3}\pi r^3$ where $r = 2.1$ cm.
2. Calculate mass using Mass = Density × Volume.

Derivation:

$$V = \frac{4}{3} \times \frac{22}{7} \times (2.1)^3 = 38.808 \text{ cm}^3. \text{ Mass} = 38.808 \times 8.9 = 345.3912 \text{ g}.$$

Surface Areas and Volumes — Volume of sphere

14. **14.** A metallic sphere of radius 6 cm is melted and recast into the shape of a cylinder of radius 4 cm. Find the height of the cylinder.

Answer:

$$18 \text{ cm}$$

Solution:

1. Equate the volume of the sphere to the volume of the cylinder.
2. Substitute the formula for the volume of a sphere $\frac{4}{3}\pi r^3$ and cylinder $\pi r^2 h$.
3. Solve for the height h.

Derivation:

$$\frac{4}{3}\pi(6)^3 = \pi(4)^2 h$$
$$\implies \frac{4}{3} \cdot 216 = 16h$$
$$\implies 4 \cdot 72 = 16h$$
$$\implies 288 = 16h$$
$$\implies h = 18 \text{ cm}$$

Surface Areas and Volumes — Conversion of solid from one shape to another

15. **15.** Two cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.

Answer:

$$160 \text{ cm}^2$$

Solution:

1. Find the edge length of the cube by taking the cube root of the volume.
2. Determine the dimensions of the new cuboid (length, breadth, height).
3. Apply the surface area formula $2(lb + bh + lh)$.

Derivation:

$$a^3 = 64$$
$$\implies a = 4 \text{ cm. New dimensions: } l = 8, b = 4, h = 4. \text{ SA} = 2(8 \cdot 4 + 4 \cdot 4 + 8 \cdot 4) = 2(32 + 16 + 32) = 2(80) = 160 \text{ cm}^2$$

Surface Areas and Volumes — Combination of solids

16. **16.** A conical tent is 10 m high and the radius of its base is 24 m. Find the slant height of the tent and the cost of the canvas required to make the tent, if the cost of 1 m^2 canvas is Rs. 70.

Answer:

$$\text{Slant height} = 26 \text{ m}, \text{Cost} = \text{ Rs. } 137280$$

Solution:

1. Calculate slant height $l = \sqrt{r^2 + h^2}$.
2. Calculate the curved surface area (CSA) of the cone using $\pi r l$.
3. Multiply the CSA by the given cost per square meter.

Derivation:

$$l = \sqrt{24^2 + 10^2} = \sqrt{576 + 100} = 26 \text{ m. CSA} = \frac{22}{7} \times 24 \times 26 = \frac{13728}{7} \text{ m}^2. \text{ Cost} = \frac{13728}{7} \times 70 = 137280$$

Surface Areas and Volumes — Cone

17. **17.** A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area.

Answer:

$$220 \text{ mm}^2$$

Solution:

1. Identify the cylindrical part length by subtracting the radii of the two hemispheres from the total length.
2. Calculate the CSA of the cylinder and the CSA of the two hemispheres.
3. Sum the two areas to find the total surface area.

Derivation:

$$r = 2.5 \text{ mm}, h_{\text{cyl}} = 14 - (2.5 + 2.5) = 9 \text{ mm. SA} = 2\pi rh + 2(2\pi r^2) = 2\pi r(h + 2r) = 2 \times \frac{22}{7} \times 2.5 \times (9 + 5) = 2 \times \frac{22}{7} \times 2.5 \times 14 = 220 \text{ mm}^2$$

Surface Areas and Volumes — Combination of solids

18. **18.** A hemispherical tank, full of water, is emptied by a pipe at the rate of $3\frac{4}{7}$ litres per second. How much time will it take to empty half the tank, if it is 3 m in diameter? (Take $\pi = \frac{22}{7}$)

Answer:

$$16.5 \text{ minutes}$$

Solution:

1. Calculate the volume of the hemispherical tank using $\frac{2}{3}\pi r^3$.
2. Find half the volume and convert it to litres ($1 \text{ m}^3 = 1000$ L).
3. Divide by the rate of flow and convert seconds to minutes.

Derivation:

$$r = 1.5 \text{ m. Vol} = \frac{2}{3} \times \frac{22}{7} \times (1.5)^3 \approx 7.07 \text{ m}^3\text{. Half Vol} = 3.535 \text{ m}^3 = 3535 \text{ L. Rate} = \frac{25}{7} \text{ L/s. Time} = 3535 \div \frac{25}{7} = 990 \text{ seconds} = 16.5 \text{ minutes.}$$

Surface Areas and Volumes — Hemisphere

19. **19.** A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 2 cm and the diameter of the base is 4 cm. If a right circular cylinder circumscribes the toy, find the difference between the volume of the cylinder and the volume of the toy. (Take $\pi = 3.14$)

Answer:

$$16.74 \text{ cm}^3$$

Solution:

1. Identify the dimensions: Cone radius $r = 2$ cm, height $h = 2$ cm. Hemisphere radius $r = 2$ cm.
2. Calculate volume of the toy: V_{toy} = V_{cone} + V_{hemisphere} $= \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$.
3. Determine dimensions of the circumscribing cylinder: Radius $r = 2$ cm, Total height $H = h + r = 2 + 2 = 4$ cm.
4. Calculate volume of the cylinder: $V_{cyl} = \pi r^2 H$.
5. Calculate the difference: $V_{cyl} - V_{toy}$.

Derivation: V_{toy} $= \frac{1}{3}(3.14)(2^2)(2) + \frac{2}{3}(3.14)(2^3) = \frac{8.37+16.74}{1} = 25.12$ cm^3.
$V_{cyl} = (3.14)(2^2)(4) = 50.24$ cm^3.
Difference $= 50.24 - 25.12 = 25.12$ cm^3.Correction: $V_{toy} = 8.373 + 16.746 = 25.12$. Difference $= 50.24 - 25.12 = 25.12$ cm^3. *Surface Areas and Volumes — Combination of solids*

20. **20.** A metallic sphere of radius 6 cm is melted and recast into the shape of a right circular cylinder of radius 6 cm. Find the height of the cylinder.

Answer:

8 cm

Solution:

1. Equate the volume of the sphere to the volume of the cylinder since the material remains constant.
2. Formula for sphere volume: $V_s = \frac{4}{3}\pi r_s^3$.
3. Formula for cylinder volume: $V_c = \pi r_c^2 h$.
4. Substitute the given values $r_s = 6$ and $r_c = 6$.
5. Solve for h.

Derivation:

$$\begin{aligned}
\frac{4}{3}\pi(6)^3 &= \pi(6)^2 h \\
\frac{4}{3} \times 216 &= 36h \\
4 \times 72 &= 36h \\
288 &= 36h \\
h &= 8 \text{ cm}
\end{aligned}$$

Surface Areas and Volumes — Conversion of solids

21. **21.** A bucket is in the form of a frustum of a cone with a capacity of 12308.8 cm^3. The radii of the top and bottom circular ends are 20 cm and 12 cm respectively. Find the height of the bucket. (Use $\pi = 3.14$)

Answer:

15 cm

Solution:

1. State the formula for the volume of a frustum: $V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$.
2. Substitute given values: $V = 12308.8$, $R = 20$, $r = 12$.
3. Calculate the term $(R^2 + r^2 + Rr) = 400 + 144 + 240 = 784$.
4. Set up the equation: $12308.8 = \frac{1}{3} \times 3.14 \times h \times 784$.
5. Solve for h.

Derivation:

$$\begin{aligned} 12308.8 &= \frac{3.14 \times h \times 784}{3} \\ 36926.4 &= 2461.76h \\ h &= \frac{36926.4}{2461.76} = 15 \text{ cm} \end{aligned}$$

Surface Areas and Volumes — Frustum of a cone

22. **22.** A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area.

Answer:

$$220 \text{ mm}^2$$

Solution:

1. Determine the dimensions: Total length = 14 mm, radius of hemisphere $r = 2.5$ mm.
2. Height of cylindrical part $h = 14 - 2.5 - 2.5 = 9$ mm.
3. Surface area of capsule = CSA of cylinder + 2× CSA of hemisphere.
4. Formula: $2\pi rh + 2(2\pi r^2) = 2\pi r(h + 2r)$.
5. Calculate value using $\pi = 22/7$.

Derivation:

$$\begin{aligned} SA &= 2 \times \frac{22}{7} \times 2.5(9 + 2(2.5)) \\ SA &= 2 \times \frac{22}{7} \times 2.5(14) \\ SA &= 2 \times 22 \times 2.5 \times 2 = 220 \text{ mm}^2 \end{aligned}$$

Surface Areas and Volumes — Combination of solids

23. **23.** If the probability of an event E occurring is 0.65, what is the probability of the event 'not E'?

Answer: (B)

Solution:

1. Recall the property of complementary events: $P(E) + P(\text{not } E) = 1$.
2. Substitute the given value $P(E) = 0.65$.
3. Solve for $P(\text{not } E) = 1 - 0.65$.

Derivation:

$$P(\bar{E}) = 1 - P(E) = 1 - 0.65 = 0.35$$

Probability — Complementary events

24. 24. The probability of winning a game is $\frac{7}{12}$. What is the probability of losing the game?

Answer: (C)

Solution:

1. The event 'losing' is the complement of 'winning'.
2. Use the formula $P(\text{losing}) = 1 - P(\text{winning})$.
3. Calculate $1 - \frac{7}{12}$.

Derivation:

$$P(\text{Losing}) = 1 - \frac{7}{12} = \frac{12 - 7}{12} = \frac{5}{12}$$

Probability — Complementary events

25. 25. A bag contains balls of different colours. If the probability of picking a red ball is 2/9, what is the probability that the ball picked is NOT red?

Answer: (D)

Solution:

1. The event of picking a ball that is not red is the complement of picking a red ball.
2. Apply $P(\text{not red}) = 1 - P(\text{red})$.
3. Calculate $1 - 2/9$.

Derivation:

$$P(\text{not Red}) = 1 - \frac{2}{9} = \frac{9 - 2}{9} = \frac{7}{9}$$

Probability — Complementary events

26. 26. If $P(E) = 0.07$, find $P(\text{not } E)$.

Answer: (A)

Solution:

1. Use the complement rule $P(\text{not } E) = 1 - P(E)$.
2. Subtract 0.07 from 1.

Derivation:

$$P(\bar{E}) = 1 - 0.07 = 0.93$$

Probability — Complementary events

27. 27. A student solves a problem correctly with a probability of 0.85. What is the probability that the student solves the problem incorrectly?

Answer: (C)

Solution:

1. The event of solving incorrectly is the complement of solving correctly.
2. Use $P(\text{incorrect}) = 1 - P(\text{correct})$.
3. Calculate $1 - 0.85$.

Derivation:

$$P(\text{incorrect}) = 1 - 0.85 = 0.15$$

Probability — Complementary events

28. 28. Assertion (A): If the probability of winning a game is 0.7, then the probability of losing the game is 0.3. Reason (R): The sum of the probabilities of all complementary events is 1.

Answer: (A)

Solution:

1. Check if the events winning and losing are complementary.
2. Check if the sum of probabilities equals 1.
3. Verify the calculation 1 - 0.7 = 0.3.

Derivation:

$$P(E) + P(\text{not E}) = 1$$
$$\implies 0.7 + 0.3 = 1.0$$

Probability — complementary events

29. 29. Assertion (A): The probability of an event happening is 0.45, then the probability of the event not happening is 0.65. Reason (R): For any event E, $P(\bar{E}) = 1 - P(E)$.

Answer: (D)

Solution:

1. Identify the formula $P(\bar{E}) = 1 - P(E)$.
2. Calculate $1 - 0.45 = 0.55$.
3. Since $0.55 \neq 0.65$, Assertion is false.

Derivation:

$$P(\bar{E}) = 1 - 0.45 = 0.55$$

Probability — complementary events

30. **30.** Assertion (A): If a fair coin is tossed, the probability of getting a head is 0.5 and the probability of not getting a head is 0.5. Reason (R): Getting a head and not getting a head are complementary events.

Answer: (A)

Solution:

1. Identify the outcomes of a coin toss as Head and Tail.
2. Understand that not getting a head is equivalent to getting a tail.
3. Verify $P(H) = 0.5$ and $P(\bar{H}) = 1 - 0.5 = 0.5$.

Derivation:

$$P(H) = 1/2 = 0.5; P(\text{not } H) = 1 - 0.5 = 0.5$$

Probability — complementary events

31. **31.** Assertion (A): If the probability of selecting a red ball from a bag is 2/7, the probability of not selecting a red ball is 5/7. Reason (R): The probability of an impossible event is 1.

Answer: (C)

Solution:

1. Check the complementary probability: $1 - 2/7 = 5/7$. Assertion is true.
2. Identify that the probability of an impossible event is 0, not 1. Reason is false.

Derivation:

$$P(\bar{E}) = 1 - 2/7 = 5/7$$

Probability — complementary events

32. **32.** Assertion (A): If $P(E) = 0.12$, then $P(\bar{E}) = 0.88$. Reason (R): $P(E)+P(\bar{E}) = 1$.

Answer: (A)

Solution:

1. Apply the formula $P(\bar{E}) = 1 - P(E)$.
2. Calculate $1 - 0.12 = 0.88$.
3. Confirm that the reason provided is the definition of complementary events.

Derivation:

$$0.12 + 0.88 = 1.00$$

Probability — complementary events

33. **33.** A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears (i) a two-digit number, (ii) a perfect square number, (iii) a number divisible by 5. Use the concept of complementary events to verify the probability of the event 'not getting a perfect square'.

Answer:

$$(i)9/10, (ii)1/10, (iii)1/5; \text{Complement} : 9/10$$

Solution:

1. Identify the total number of outcomes as 90.
2. List perfect squares between 1 and 90: {1, 4, 9, 16, 25, 36, 49, 64, 81}. Total = 9.
3. Calculate P(Perfect Square) = 9/90 = 1/10.
4. Use the complementary event formula: P(Not E) = 1 - P(E).
5. Calculate P(Not Perfect Square) = $1 - 1/10 = 9/10$.

Derivation:

$$P(E) = \frac{9}{90} = \frac{1}{10}$$
$$P(\bar{E}) = 1 - P(E)$$
$$P(\bar{E}) = 1 - \frac{1}{10} = \frac{9}{10}$$

Probability — Complementary events

34. **34.** A game consists of tossing a one-rupee coin 3 times and noting its outcome each time. Hanif wins if all the tosses give the same result i.e., three heads or three tails, and loses otherwise. Calculate the probability that Hanif will lose the game using the complementary event property.

Answer:

$$3/4$$

Solution:

1. List all possible outcomes: {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}. Total = 8.
2. Identify favorable outcomes for winning: {HHH, TTT}. Count = 2.
3. Calculate P(Win) = 2/8 = 1/4.
4. Identify the event of losing as the complement of winning.
5. Apply P(Lose) = $1 - P(Win)$.

Derivation: P(Win) = $\frac{2}{8} = \frac{1}{4}$
P(Lose) = 1 - P(Win) \\ P(Lose) = 1 - $\frac{1}{4} = \frac{3}{4}$ *Probability — Complementary events*

35. **35.** A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is double that of a red ball, find the number of blue balls in the bag. Further, if a ball is drawn at random, what is the probability of not drawing a red ball?

Answer:

$$10\text{blue balls; Probability of not red} = 2/3$$

Solution:

1. Let the number of blue balls be x. Total balls = 5 + x.
2. $P(Red) = 5/(5+x), P(\text{Blue}) = x/(5+x)$.
3. Given P(Blue) = 2 * P(Red), solve for x: x/(5+x) = 2 * (5/(5+x)).
4. Solving results in x = 10.
5. Calculate P(Not Red) = $1 - P(Red)$.

Derivation: $\frac{x}{5+x} = 2 \times \frac{5}{5+x} \implies x = 10$
$P(\text{Red}) = \frac{5}{15} = \frac{1}{3}$
$P(\text{Not Red}) = 1 - \frac{1}{3} = \frac{2}{3}$ *Probability — Complementary events*

36. **36.** Two dice are thrown simultaneously. Find the probability of getting (i) a sum of 8, (ii) a doublet, (iii) a sum greater than 9. Use the complementary event concept to find the probability of getting a sum not equal to 8.

Answer:

$$(i)5/36, (ii)6/36, (iii)6/36; \text{Complement} : 31/36$$

Solution:

1. Total outcomes for two dice = $6 * 6 = 36$.
2. Identify outcomes for sum 8: {(2,6), (3,5), (4,4), (5,3), (6,2)}. Count = 5.
3. $P(Sum8) = 5/36$.
4. Use complement property for 'Sum not 8': P(Not 8) = 1 - P(8).
5. Final calculation: 1 - 5/36 = 31/36.

Derivation: P(Sum=8) = $\frac{5}{36}$
$P(\text{Sum} \neq 8) = 1 - P(\text{Sum=8})$ \\ P(Sum $\neq$ 8) $= 1 - \frac{5}{36} = \frac{31}{36}$ *Probability — Complementary events*

37. **37.** In $\triangle ABC$, a circle is inscribed touching sides AB, BC, and CA at points P, Q, and R respectively. If $AB = 12$ cm, $BC = 8$ cm, and $AC = 10$ cm, find the length of AP.

Answer:

$$7cm$$

Solution:

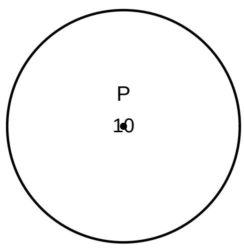
1. Let $AP = AR = x$, $BP = BQ = y$, and $CQ = CR = z$.
2. Set up the system of equations based on side lengths: $x + y = 12$, $y + z = 8$, $z + x = 10$. Solve for x.

Derivation:

$$x + y = 12, \ y + z = 8, \ z + x = 10$$
$$(x + y + z) = (12 + 8 + 10)/2 = 15$$
$$x = 15 - (y + z) = 15 - 8 = 7$$

Circles — Tangents to a circle

38. **38.** From a point P, 10 cm away from the centre of a circle, the length of the tangent drawn to the circle is 8 cm. Find the radius of the circle.



Answer:

$$6cm$$

Solution:

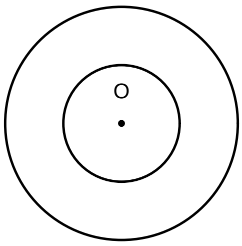
1. Use the property that the tangent is perpendicular to the radius at the point of contact.
2. Apply Pythagoras theorem in the right-angled triangle formed by the radius, tangent, and the line from the centre to P.

Derivation:

$$\begin{aligned} r^2 + 8^2 &= 10^2 \\ r^2 + 64 &= 100 \\ r^2 &= 36 \\ r &= 6 \text{ cm} \end{aligned}$$

Circles — Tangents to a circle

39. **39.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.



Answer:

$$8cm$$

Solution:

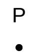
1. Let O be the centre, AB be the chord of the larger circle tangent to the smaller circle at M.
2. In $\triangle OMA$, use Pythagoras theorem to find AM, then $AB = 2 \times AM$.

Derivation:

$$\begin{aligned} OM &= 3,\ OA = 5 \\ AM &= \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = 4 \\ AB &= 2 \times 4 = 8 \text{ cm} \end{aligned}$$

Circles — Tangents to a circle

40. **40.** If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of $80°$, find $\angle POA$.



Answer:

$$50^\circ$$

Solution:

1. The line OP bisects the angle $\angle APB$. Thus $\angle APO = 40^\circ$.
2. In $\triangle OAP$, $\angle OAP = 90^\circ$. Therefore $\angle POA = 90^\circ - 40^\circ$.

Derivation:

$$\begin{aligned} \angle APB &= 80^\circ \\ \Rightarrow \angle APO &= 40^\circ \\ \angle OAP &= 90^\circ \\ \angle POA = 180^\circ - 90^\circ - 40^\circ &= 50^\circ \end{aligned}$$

Circles — Tangents to a circle